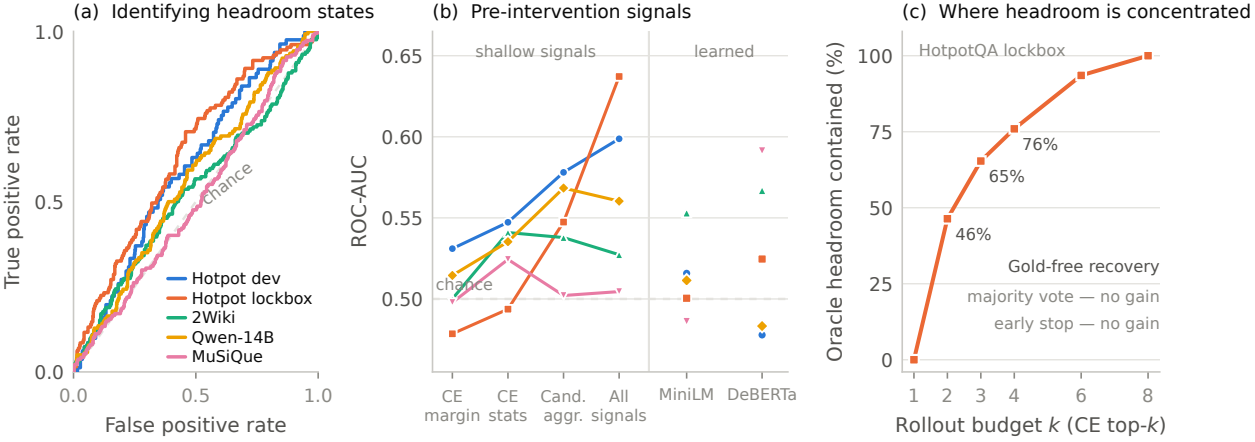
(a) Identifying headroom states
True positive rate
False positive rate
1.0
0.5
0.0
0.0
0.5
1.0
Chance
Hotpot dev
Hotpot lockbox
2Wiki
Qwen-14B
MuSiQue
(b) Pre-intervention signals
ROC-AUC
0.65
0.60
0.55
0.50
shallow signals
learned
chance
CE margin
CE stats
Cand. aggr.
All signals
MiniLM
DeBERTa
(c) Where headroom is concentrated
Oracle headroom contained (%)
100
75
50
25
0
HotpotQA lockbox
76%
65%
46%
Gold-free recovery
majority vote — no gain
early stop — no gain
1
2
3
4
6
8
Rollout budget k (CE top-k)